Networking in AWS Cloud

Lesson

Skills you will learn

Cloud Computing: Compute & Networking

Lesson Overview ✏️

Virtual Private Cloud enables users to create isolated network environments within AWS. This lesson will provide in-depth awareness of the components, functionalities, and security measures of VPCs.

Learning Objectives 📓

By the end of this lesson, you will be able to:

1. Understand the purpose and structure of AWS VPCs.
2. Learn about the roles of public and private subnets, route tables, internet gateways, and NAT gateways within a VPC.
3. Explore VPC security features, including network ACLs, security groups, and VPC flow logs.
4. Gain insights into additional VPC features such as Elastic IP addresses, Elastic Network Interfaces, and VPC peering.

The Lesson 🔑

AWS VPC (Virtual Private Cloud)

Amazon VPC (Virtual Private Cloud) provides an isolated space within the AWS Infrastructure which serves as your own network environment within the cloud for which you can have your own network subnets and IP Address allocations.

IP allocations are for both IPv4 and IPv6.

By default, you have a default VPC when your account is created to allow you to launch public instances. If you delete the default VPC you can just as easily create a new VPC.

Ask AI Tutor

CIDR Block	Reason
10.0.0.0	Network Address
10.0.0.1	AWS reserved this for VPC router address
10.0.0.2	IP address of DNS server is always set to base VPC network address range plus two; VPC also reserves base of each subnet range plus two
10.0.0.2	AWS reserved for future use
10.0.0.255	Network broadcast address. Broadcast in the VPC is not supported thus this address is reserved.

VPC Subnets

VPC Subnets allow you to form a subset of IP address ranges you can use within your VPC. Within the VPC you are able to provision your resources. Each subnet within our VPC can reside in one availability zone, with availability zones containing more than one subnets at a time.

There are generally two subnets:

1. Public Subnet
2. Private Subnet

Public Subnet

Public subnets allow for resources in our VPC to communicate directly with external networks (Internet) and this is made possible because of a few components.

1. Route Tables
2. Internet Gateway
3. Public IP Address is an IP address assigned to your resources in the VPC that can be reached from the Internet.

Route Tables

Route Tables are rules that are used by resources to direct traffic in and out of the VPC.

For traffic within the VPC, it will be directed locally but for external traffic, that will be done using an Internet Gateway.

Internet Gateway

Internet Gateway is a feature of the VPC that allows communication between the Internet and the resources found within your VPC. You specify this as a target in route table when accessing the Internet from your VPC.

The Internet Gateway allows you route traffic to the internet and also protects your IP address from being exposed to the Internet by serving as a NAT (Network Address Translation) service.

NAT Gateway

NAT Gateway is used to map private IP address to a public IP address and thereby allows private subnets to connect to the Internet.

If you are connecting a private subnet to the internet using the NAT gateway, the private route table sends all internet facing traffic to the NAT gateway and from there the public route table assigns all Internet traffic to the Internet gateway.

This connection prevents external traffic from connecting with your private instances.

Private Subnets

Private subnets do not allow for direct communication between your resources and the Internet. Traffic stays within the private network (Your VPC) using the Private IP Address.

Resources that do not need to communicate with the Internet are mostly placed within the Private Subnet to serve as an extra layer of security example your databases and your backend applications.

VPC Security

There are multiple security measures available when working with your VPC and there are some general recommendations;

- Employ network ACLs and security groups
- Use IAM policies to limit access to your resources
- Implement the use of VPC Flow logs to review VPC traffic
- Employ multiple Availability Zones where possible

Network ACL (Access Control Lists)

A network ACL is an optional layer of security that serves as a stateless firewall controlling traffic in and out of one or more subnets.

Just like the default VPC, there is a default ACL available which allows all traffic both inbound and outbound.

The rule number determines which rule to follow first starting with the lowest.

Security Groups

A security group is a virtual firewall which works on the resource level to control traffic inbound and outbound of resources in a stateful manner.

By default, security groups allows all outbound traffic and blocks inbound traffic.

In addition for your resources, you can use chaining of security groups to ensure that only specific security groups can access resources in another security group.

Network ACLs Vs Security Groups

Network ACLs	Security Groups
Work on the subnet level of VPC	Work at the instance level and associated with a static network interface
Is a stateless firewall	Is a stateful firewall
Applies to all instances within the associated subnet.	Applies to the instance it is associated with
Supports allow and deny rules	Supports allow rules only
Rules processed in order of rule number	Evaluates all rules before deciding to allow traffic or not.

VPC Flow

VPC Flow Logs are used to capture traffic information to and from VPC network Interfaces. The captured IP traffic can be sent to other AWS services like the S3 bucket or CloudWatch logs to be analyzed.

VPC flow logs help in the following ways:

- Helps determine direction of traffic to and from network interfaces.
- For traffic getting to your instances, VPC Flow logs is able to monitor that.
- Helps to diagnose rules that are restrictive within your security groups.

Additional VPC Features

1. Elastic IP address is a static, public IPv4 address that can be associated with any instance or network within your VPC . The elastic IP address can be easily mapped to another instance if there is a failure.
2. Elastic Network Interface is a virtual networking component within your VPC that represents a virtual network card. This can be moved across resources in the same availability zone and it maintains its private IP address, elastic IP address and MAC address.
3. VPC Peering is a feature within the AWS VPC which allows you to have a one-to-one connection two VPCs across the same account or multiple accounts and across the same region or different regions. Once you create a peering connection you should update your route tables to reflect the details of the peering connection.

AI TUTOR ACTIONS

Generate a practice quiz Summarize this unit Ask about this unit

Complete unit